Obelisk Proof Assistant — Reference Demonstration

Variant: Higher-Order Logic & Morse–Kelley Class Theory • Author: Georgy Dunaev • contact@obelisk.mozmail.com

1) Propositional logic

1.01) Axm. A01 (Affirmation of Consequent):

$(\alpha \to (\beta \to \alpha))$

Syntax side condition: α, β : Bool

1.02) Axm. A02 (Distribution of Implication):

$((\alpha \to (\beta \to \gamma)) \to ((\alpha \to \beta) \to (\alpha \to \gamma)))$

Syntax side condition: α, β, γ : Bool

1.03) Axm. A03 (Conjunction Elimination Left):

$((\alpha \wedge \beta) \to \alpha)$

Syntax side condition: α, β : Bool

1.04) Axm. A04 (Conjunction Elimination Right):

$((\alpha \wedge \beta) \to \beta)$

Syntax side condition: α, β : Bool

1.05) Axm. A05 (Conjunction Introduction):

$(\alpha \to (\beta \to (\alpha \wedge \beta)))$

Syntax side condition: α, β : Bool

1.06) Axm. A06 (Disjunction Introduction Left):

$(\alpha \to (\alpha \vee \beta))$

Syntax side condition: α, β : Bool

1.07) Axm. A07 (Disjunction Introduction Right):

$(\beta \to (\alpha \vee \beta))$

Syntax side condition: α, β : Bool

1.08) Axm. A08 (Disjunction Elimination / Cases):

$((\alpha \to \gamma) \to ((\beta \to \gamma) \to ((\alpha \vee \beta) \to \gamma)))$

Syntax side condition: α, β, γ : Bool

1.09) Axm. A09 (Negation Introduction / Contradiction):

$((\neg\alpha) \to (\alpha \to \bot))$

Syntax side condition: α : Bool

1.10) Axm. A10 (Reductio Ad Absurdum):

$((\alpha \to \beta) \to ((\alpha \to \neg\beta) \to \neg\alpha))$

Syntax side condition: α, β : Bool

1.11) Axm. A11 (Ex Falso Quodlibet):

$(\bot \to \alpha)$

Syntax side condition: α : Bool

1.12) Axm. A12 (Law of Excluded Middle):

$(\alpha \vee \neg\alpha)$

Syntax side condition: α : Bool

1.13) Rule Modus Ponens (MP):

$\alpha, (\alpha \to \beta) \vdash \beta$

Syntax side condition: α, β : Bool

2) Predicate logic

2.01) Axm. Discourse domain is non-empty:

$\exists(K : \mathrm{Class})\ \top$

Syntax side condition: K : Class

2.02) Rule Forall Instantiation (FI):

$((\forall(x : \mathrm{Class})\ \varphi) \to \varphi[t/x])$

Syntax side condition: FFI(t, v_x, φ)

2.03) Rule Exists Instantiation (EI):

$(\varphi[t/x] \to \exists(x : \mathrm{Class})\ \varphi)$

Syntax side condition: $\mathrm{FFI}(t, v_x, \varphi)$

2.04) Rule Forall Abstraction (FA / Bernays):

$(\psi \to \varphi) \vdash (\psi \to \forall(x : \mathrm{Class})\ \varphi)$

Syntax side condition: $(\mathrm{NFV}(v_x, \psi) \land \mathrm{NFV}(v_x, \Gamma))$

2.05) Rule Exists Abstraction (EA / Bernays):

$(\varphi \to \psi) \vdash (\exists(x : \mathrm{Class})\ (\varphi \to \psi))$

Syntax side condition: $(\mathrm{NFV}(v_x, \psi) \land \mathrm{NFV}(v_x, \Gamma))$

2.06) Rule Generalisation (Gen):

$\varphi \vdash \forall(x : \mathrm{Class})\ \varphi$

Syntax side condition: $\mathrm{NFV}(v_x, \Gamma)$

3) Set/class-theoretic foundations of mathematics

3.01) Axm. Class Comprehension Axiom:

$\forall(x : \mathrm{Class})\ ((x \in \{\ y \mid \varphi(y)\ \}) \longleftrightarrow (\mathrm{isSet}(x) \land \varphi(x)))$

Syntax side condition: $\mathrm{NFV}(v_C, \varphi)$

3.02) Axm. Extensionality Axiom:

$\forall(A, B : \mathrm{Class})\ ((\forall(x : \mathrm{Class})\ ((x \in A) \longleftrightarrow (x \in B))) \to (A = B))$

Syntax side condition: $A, B, x : \mathrm{Class}$

3.08) Axm. Heredity Axiom (Subclasses of Sets are Sets):

$\forall(A, b : \mathrm{Class})\ ((\mathrm{isSet}(b) \land (A \subseteq b)) \to \mathrm{isSet}(A))$

Syntax side condition: $A, b : \mathrm{Class}$

3.12) Axm. Regularity (Foundation) Axiom:

$\forall(A : \mathrm{Class})\ ((A \neq \varnothing) \to \exists(x : \mathrm{Class})\ ((x \in A) \land ((x \cap A) = \varnothing)))$

Syntax side condition: $A, x : \mathrm{Class}$

3.13) Axm. Pairing Axiom:

$\forall(a, b : \mathrm{Class})\ ((\mathrm{isSet}(a) \land \mathrm{isSet}(b)) \to \mathrm{isSet}(\{a, b\}))$

Syntax side condition: $a, b : \mathrm{Class}$

3.14) Axm. Union Axiom:

$\forall(a : \mathrm{Class})\ (\mathrm{isSet}(a) \to \mathrm{isSet}(\bigcup a))$

Syntax side condition: $a : \mathrm{Class}$

3.22) Def. IF-THEN-ELSE-FI (Class conditional with NFV side condition and context turnstile):

$W, \Gamma \vdash (\llbracket \mathrm{IF}\ \psi\ \mathrm{THEN}\ A\ \mathrm{ELSE}\ B\ \mathrm{FI} \rrbracket = \{\ v \mid ((\psi \land (v \in A)) \lor (\neg\psi \land (v \in B)))\ \})$

Syntax side condition: $(\mathrm{NFV}(v_v, \psi) \land \mathrm{NFV}(v_v, \Gamma))$

3.23) Def. simple_if (metalevel conditional over Type_Expression):

$\forall(T_e : \mathrm{Type_Expression}),\ \triangleleft\ (\mathrm{simple_if}(\llbracket T_e \rrbracket) : (\mathrm{Bool} \to (\llbracket T_e \rrbracket \to (\llbracket T_e \rrbracket \to \llbracket T_e \rrbracket))))\ \triangleright$

$\forall(T_e : \mathrm{Type_Expression}),\ \triangleleft\ \forall(t_1 : \llbracket T_e \rrbracket)\ \forall(t_2 : \llbracket T_e \rrbracket)\ ((\mathrm{simple_if}(\llbracket T_e \rrbracket, \mathrm{true}, t_1, t_2) = t_1) \land (\mathrm{simple_if}(\llbracket T_e \rrbracket, \mathrm{false}, t_1, t_2) = t_2))\ \triangleright$

Syntax side condition: $T_e : \mathrm{Type_Expression}, t_1 : \llbracket T_e \rrbracket, t_2 : \llbracket T_e \rrbracket$

3.24) Def. if_dep (dependent conditional over Bool → Type_Expression):

$\forall(T_e : (\mathrm{Bool} \to \mathrm{Type_Expression})),\ \triangleleft\ (\mathrm{if_dep} : \Pi(b : \mathrm{Bool}),\ \llbracket T_e(\mathrm{true}) \rrbracket \to (\llbracket T_e(\mathrm{false}) \rrbracket \to \llbracket T_e(b) \rrbracket))\ \triangleright$

$\forall(T_e : (\mathrm{Bool} \to \mathrm{Type_Expression})),\ \triangleleft\ \forall(b : \mathrm{Bool})\ \forall(t_1 : \llbracket T_e(\mathrm{true}) \rrbracket)\ \forall(t_2 : \llbracket T_e(\mathrm{false}) \rrbracket)\ ((\mathrm{if_dep}(b, t_1, t_2) : \llbracket T_e(b) \rrbracket) \land ((\mathrm{if_dep}(\mathrm{true}, t_1, t_2) = t_1) \land (\mathrm{if_dep}(\mathrm{false}, t_1, t_2) = t_2)))\ \triangleright$

Syntax side condition: $T_e : (\mathrm{Bool} \to \mathrm{Type_Expression}), b : \mathrm{Bool}, t_1 : \llbracket T_e(\mathrm{true}) \rrbracket, t_2 : \llbracket T_e(\mathrm{false}) \rrbracket$

3.32) Thm. Ordinal Induction (∈-Ind):

$((\forall(x : \mathrm{Class})\ ((x \in \mathrm{On}) \to ((\forall(y : \mathrm{Class})\ ((y \in x) \to \varphi(y))) \to \varphi(x)))) \to \forall(x : \mathrm{Class})\ ((x \in \mathrm{On}) \to \varphi(x)))$

Syntax side condition: $(\mathrm{Fresh}(v_y, v_x) \land \mathrm{NFV}(v_y, \varphi(x)))$

3.33) Def. Power Class Definition (□(A)):

$\forall(A, x : \mathrm{Class})\ ((x \in \square(A)) \longleftrightarrow (\mathrm{isSet}(x) \land (x \subseteq A)))$

Syntax side condition: $\mathrm{Fresh}(v_x, v_A)$

3.34) Axm. Power Set Axiom (Power Class of a Set is a Set):

$\forall(a : \mathrm{Class})\ (\mathrm{isSet}(a) \to \mathrm{isSet}(\square(a)))$

Syntax side condition: $a : \mathrm{Class}$

3.37) Axm. Replacement / Limitation of Size Axiom:
∀(F, a : Class) ((isClassFunction(F) ∧ isSet(a)) → isSet(F[a]))
Syntax side condition: F, a : Class

3.40) Def. Dunaev Tagged Ordered Pair:
∀(A, B : Class) ([A, B]_D = { z | ∃(a ∈ A) (z = {{{{a}}}, {∅}}) ∨ ∃(b ∈ B) (z = {{{{b}}}, {∅, {∅}}}) })
Syntax side condition: A, B : Class

3.41) Thm. First Projection of Dunaev Tagged Ordered Pair:
∀(A, B : Class) ({ a | ({{{{a}}}, {∅}} ∈ [A, B]_D) } = A)
Syntax side condition: A, B : Class

3.42) Thm. Second Projection of Dunaev Tagged Ordered Pair:
∀(A, B : Class) ({ b | ({{{{b}}}, {∅, {∅}}} ∈ [A, B]_D) } = B)
Syntax side condition: A, B : Class

3.45) Axm. Infinity Axiom:
∃(x : Class) (isSet(x) ∧ ((∅ ∈ x) ∧ ∀(y : Class) ((y ∈ x) → ((y ∪ {y}) ∈ x))))
Syntax side condition: x, y : Class

3.46) Axm. Global Choice Axiom:
∃(F : Class) (isClassFunction(F) ∧ ∀(x : Class) ((isSet(x) ∧ (x ≠ ∅)) → (F(x) ∈ x)))
Syntax side condition: F, x : Class

3.47) Def. Universal Class (V):
(V = { x | (x = x) })
Syntax side condition: V : Class

3.53) Def. Transitive Class Predicate:
(Tr(A) ⟷ ∀(x : Class) ((x ∈ A) → (x ⊆ A)))
Syntax side condition: A, x : Class

3.66) Axm. Tarski's Axiom of Universes:
∀(x : Class) (isSet(x) → ∃(u : Class) ((isUniverse(u) ∧ (x ∈ u)) ∧ isSet(u)))
Syntax side condition: x, u : Class

3.67) Def. Semi-Singleton Class Predicate (semi_singleton):
(semi_singleton(X) ⟷ ∀(u : Class) ∀(v : Class) (((u ∈ X) ∧ (v ∈ X)) → (u = v)))
Syntax side condition: X, u, v : Class

3.68) Def. Value Extraction via Semi-Singleton Subclasses (sss / Value):
(sss(X) = ⋃({ z | ((z ⊆ X) ∧ semi_singleton(z)) }))
Syntax side condition: X, z : Class

3.69) Thm. Theorem of Evaluation of a Family ((Λx, t)(z) = t[z/x]):
∀(z : Class) ((isSet(z) ∧ isSet(t[z/x])) → (((Λ(x : Class), t))(z) = t[z/x]))
Syntax side condition: (t : Class.Expression), (v_x : Variable), (z : Class)

3.70) Def. Set-Theoretic Model of Dependent Sum (ΣB / Sigma Model):
(ΣB = { p | ∃(x : Class) ∃(y : Class) (((x ∈ Dom(B)) ∧ (y ∈ B(x))) ∧ (p = [x, y]_D)) })
Syntax side condition: B, p, x, y : Class

3.71) Def. Set-Theoretic Model of Dependent Product (ΠB / Pi Model):
(ΠB = { f | ((isClassFunction(f) ∧ (Dom(f) = Dom(B))) ∧ ∀(x : Class) ((x ∈ Dom(B)) → (f(x) ∈ B(x)))) })
Syntax side condition: B, f, x : Class

4) Metalogical laws

4.01) Axm. Instantiation Axiom:
∀(T_e : Type_Expression), ◁ ∀(G : Ctx) ∀(φ : Fm) ∀(t : ⟦T_e⟧) ∀(v : Variable) ((W, G ⊢ (∀(v : ⟦T_e⟧) φ)) → (W, G ⊢ (let v = t in φ))) ▷
Syntax side condition: T_e : Type_Expression, t : ⟦T_e⟧, v : Variable

4.02) Axm. Generalisation Axiom:
∀(T_e : Type_Expression), ◁ ∀(Γ : Ctx) ∀(φ : Fm) ∀(v : Variable) (¬(Enc(v) ∈ Ctx.FV(Γ)) → ((W, Γ ⊢ φ) → (W, Γ ⊢ ∀(v : ⟦T_e⟧) φ))) ▷
Syntax side condition: NFV(v, Γ), T_e : Type_Expression

4.03) Rule CoNecessitation:
(F, []) ⊢ ‹F, Γ ⊢ φ›) ⊢ (F, Γ ⊢ φ)
Syntax side condition: F : Theory, Γ : Ctx, φ : Fm

4.04) Def. Quantifier Unique (!):
∀(φ : Fm) ((!(z : Class) φ) ⟷ ∀(x, y : Class) ((φ[x/z] ∧ φ[y/z]) → (x = y)))
Syntax side condition: z, x, y : Variable

4.05) Def. Quantifier Exists and Unique (∃!):
∀(φ : Fm) (T, [] ⊢ ((∃!(x : Class) φ) ⟷ ((∃(x : Class) φ) ∧ !(x : Class) φ)))
Syntax side condition: x : Variable, φ : Fm

4.06) Axm. Class Comprehension Axiom (Metatheory):
∀(φ : Fm) ∀(x : Name) ∀(k : Class.Expr) (FFI(k, x, φ) → (T, [] ⊢ ((k ∈ { x | φ }) ⟷ (isSet(k) ∧ φ[k/x]))))
Syntax side condition: FFI(k, x, φ)

4.07) Axm. Soundness / Reflection Axiom:
∀(φ : Fm) (⟦φ⟧ → (F, [] ⊢ φ))
Syntax side condition: φ : Fm, F : Theory

4.08) Axm. Theory Inclusion (T ⊆ F):
∀(Γ : Ctx) ∀(φ : Fm) ((T, Γ ⊢ φ) → (F, Γ ⊢ φ))
Syntax side condition: T, F : Theory, Γ : Ctx, φ : Fm

4.09) Def. Syntactic Encoding Operator (Enc):
∀(T_e : Type_Expression), ◁ ∀(s : ⟦T_e⟧) (isSet(Enc(s)) ∧ (Enc(s) ∈ Emb(⟦T_e⟧))) ▷
Syntax side condition: T_e : Type_Expression, s : ⟦T_e⟧

4.10) Def. Syntactic Decoding & Roundtrip Law (Dec):
∀(T_e : Type_Expression), ◁ ∀(s : ⟦T_e⟧) ((Dec(Enc(s), Emb(⟦T_e⟧)) : ⟦T_e⟧) = (s : ⟦T_e⟧)) ▷
Syntax side condition: T_e : Type_Expression, s : ⟦T_e⟧

4.11) Def. Dual Type Embedding Operator (Emb):
∀(X : Class) ((Emb(Class)(X) ⟷ ⊤) ∧ (Emb(Set)(X) ⟷ isSet(X)))
Syntax side condition: X : Class

4.12) Def. Proof-Object Class Family (ProofsOf):
∀(s : Stmt) ∀(p : Prf) ((Enc(p) ∈ ProofsOf(Enc(s))) ⟷ CheckProof(p, s))
Syntax side condition: s : Stmt, p : Prf

4.13) Def. Class of Provable Statements (Theorems):
∀(s : Stmt) ((Enc(s) ∈ Theorems) ⟷ (∃(p : Class) (p ∈ ProofsOf(Enc(s)))))
Syntax side condition: s : Stmt, p : Class

5) Tactics and inference rules

5.01) Thm. Elementary Theorems:
((F, [(b ∈ X)]) ⊢ ∃(X : Class) (b ∈ X)) ∧ (F, [(b ∈ X)] ⊢ isSet(b)))
Syntax side condition: b, X : Class

5.02) Rule Definition to Introduction Rule:
(F, [(b ∈ X)]) ⊢ ∃(X : Class) (b ∈ X)) ⊢ (F, [(b ∈ X)] ⊢ isSet(b))
Syntax side condition: b, X : Class

5.03) Rule Deduction Rule Application:
(F, [] ⊢ ((b ∈ X) → isSet(b)))
Syntax side condition: b, X : Class

5.04) Thm. Tactics Proof Example:
F, [] ⊢ ((∃(Y : Class) ∀(x : Class) (x ∈ Y)) → ∀(x : Class) ∃(Y : Class) (x ∈ Y))
Syntax side condition: x, Y : Class

6) Incompleteness-related statements

6.01) Axm. Theories (Metalevel Stratification):
((W ⊆ Q) ∧ ((Q ⊆ T) ∧ (T ⊆ F)))
Syntax side condition: W, Q, T, F : Theory

6.02) Thm. Löb's Theorem:

∀(φ : Fm) (Q, [] ⊢ ((T, [] ⊢ ((T, [] ⊢ φ) → φ)) → (T, [] ⊢ φ)))

Syntax side condition: φ : Fm

6.03) Thm. Gödel's Second Incompleteness Theorem:

Q, [] ⊢ (¬(T, [] ⊢ ⊥) → ¬(T, [] ⊢ ¬(T, [] ⊢ ⊥)))

Syntax side condition: T, Q : Theory

Appendix: NBG Formulas-as-Types Standard Library

24 closed proof terms compiled by NBGSurfaceCompiler in Poly/ML. A formula is a type; a proof program is a term inhabiting it. The Dunaev pair injectivity theorem is the class-correct replacement for Kuratowski pairing on proper classes.

Theorem	Formula (Type)	AST	Norm
empty_class_no_elements	(∀(z : Class) ((z ∈ { x | ⊥ }) → ⊥))	7	7
empty_class_is_subclass	(∀(X : Class) (∀(z : Class) ((z ∈ { x | ⊥ }) → (z ∈ X))))	8	8
universal_class_characterization	(∀(z : Class) (((z ∈ { x | ⊤ }) → isSet(z)) ∧ (isSet(z) → (z ∈ { x | ⊤ }))))	15	15
russell_class_is_not_a_set	(isSet({ x | ((x ∈ x) → ⊥) }) → ⊥)	19	32
binary_inter_comm	(∀(A : Class) (∀(B : Class) (∀(z : Class) ((z ∈ { x | ((x ∈ A) ∧ (x ∈ B)) }) → (z ∈ { x | ((x ∈ B) ∧ (x ∈ A)) })))))	26	26
binary_union_comm	(∀(A : Class) (∀(B : Class) (∀(z : Class) ((z ∈ { x | ((x ∈ A) ∨ (x ∈ B)) }) → (z ∈ { x | ((x ∈ B) ∨ (x ∈ A)) })))))	23	23
class_diff_subclass_left	(∀(A : Class) (∀(B : Class) (∀(z : Class) ((z ∈ { x | ((x ∈ A) ∧ ((x ∈ B) → ⊥)) }) → (z ∈ A)))))	10	10
class_eq_sym	(∀(X : Class) (∀(Y : Class) ((X = Y) → (Y = X))))	6	6
class_eq_trans	(∀(X : Class) (∀(Y : Class) (∀(Z : Class) (((X = Y) ∧ (Y = Z)) → (X = Z)))))	9	9
quantifier_swap	((∃(y : Set) (∀(x : Set) (x ∈ y))) → (∀(x : Set) (∃(y : Set) (x ∈ y))))	7	7
dunaev_tagged_pair_projections	(∀(A : Class) (∀(B : Class) ((π₁([A, B]_D) = A) ∧ (π₂([A, B]_D) = B))))	3	3
cantor_diagonal_theorem	(∀(R : Class) (∀(c : Set) (isSet(c) → ((((c ∈ { x | ([x, x]_D ∈ R) → ⊥) }) → ([c, c]_D ∈ R)) ∧ ([c, c]_D ∈ R) → (c ∈ { x | ([x, x]_D ∈ R) → ⊥) })) → ⊥))))	28	50
self_in_singleton	(∀(x : Class) (isSet(x) → (x ∈ { u | (u = x) })))	8	8
self_in_succ	(∀(x : Class) (isSet(x) → (x ∈ { u | ((u ∈ x) ∨ (u = x)) })))	9	9
class_sub_succ	(∀(x : Class) (∀(z : Class) ((z ∈ x) → (z ∈ { u | ((u ∈ x) ∨ (u = x)) }))))	12	12
dunaev_pair_injective	(∀(A : Class) (∀(B : Class) (∀(C : Class) (∀(D : Class) (([A, B]_D = [C, D]_D) → ((A = C) ∧ (B = D)))))))	26	26
dunaev_cartesian_elim	(∀(A : Class) (∀(B : Class) (∀(p : Class) ((p ∈ { q | (∃(u : Set) (∃(v : Set) (((u ∈ A) ∧ (v ∈ B)) ∧ (q = [u, v]_D)))) }) → (∃(u : Set) (∃(v : Set) (((u ∈ A) ∧ (v ∈ B)) ∧ (p = [u, v]_D))))))))	9	9
inter_sub_left	(∀(A : Class) (∀(B : Class) (∀(z : Class) ((z ∈ { u | ((u ∈ A) ∧ (u ∈ B)) }) → (z ∈ A)))))	10	10
inter_sub_right	(∀(A : Class) (∀(B : Class) (∀(z : Class) ((z ∈ { u | ((u ∈ A) ∧ (u ∈ B)) }) → (z ∈ B)))))	10	10
union_sub_left	(∀(A : Class) (∀(B : Class) (∀(z : Class) ((z ∈ A) → (z ∈ { u | ((u ∈ A) ∨ (u ∈ B)) })))))	13	13
semi_singleton_sub_elim	(∀(X : Class) (∀(z : Class) ((z ∈ { w | ((w ∈ X) ∧ (∀(u : Class) (∀(v : Class) (((u ∈ X) ∧ (v ∈ X)) → (u = v))))) }) → (z ∈ X))))	9	9
family_fiber_char	(∀(B : Class) (∀(x : Class) (∀(y : Class) ((y ∈ { w | ([x, w]_D ∈ B) }) → ([x, y]_D ∈ B)))))	8	8
sigma_dependent_sum_intro	(∀(A : Class) (∀(B : Class) (∀(x : Class) (∀(y : Class) (((x ∈ A) ∧ ([x, y]_D ∈ B)) → ((x ∈ A) ∧ ([x, y]_D ∈ B)))))))	8	8
pi_dependent_product_elim	(∀(A : Class) (∀(B : Class) (∀(f : Class) (((∀(x : Class) ((x ∈ A) → ([x, f]_D ∈ B))) ∧ (∀(x : Class) (x ∈ A))) → (∀(x : Class) ([x, f]_D ∈ B))))))	10	10

Reproduce locally: python3 obelisk_cli.py compile examples/nbg_verified_programs.nbg